



An Effective Framework for Improving Performance of Daily Streamflow Estimation Using Statistical Methods Coupled with Artificial Neural Network

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Abstract—This study presents an effective framework that combines artificial neural network (ANN) and statistical methods to more efficiently, consistently, and reliably estimate the daily streamflow in ungauged basins. First, two statistical methods, including drainage area ratio (DAR) and standardization with mean (SM), are used to transfer hydrological data from gauged (donor) to ungauged (target) basins, which is known as the regionalization process. Second, to get better estimation performance, an ensemble approach is applied, which is mainly based on a weighted combination of DAR and SM. Finally, a successful strategy with an optimized ANN structure is built using daily areal precipitation for the target basin, the daily streamflow of the selected donor basin, and the estimated daily streamflow for the target basin from the best-fit method as model inputs. Its performance is tested in a case study from the Coruh River Basin, Turkey, that involved using datasets from seven streamflow gauging stations on the mainstream of Coruh River. The proposed approach has indicated the best performance on both training and testing sets. The proposed approach proves to be one of the best available practical solutions in the streamflow estimation for ungauged basins.

Keywords: Artificial neural network, ensemble approach, streamflow estimation, drainage area ratio, performance-based weighting, regionalization, ungauged basins.

1. Introduction

Streamflow is an important hydrological component of the water cycle, not only for accurate water resources planning and development but also for modeling in response to global changes such as

climate change and rapid urbanization. However, due to economic, technical, and hydromorphological constraints, continuous streamflow records are not available in many parts of the world. Therefore, streamflow estimation in ungauged basins poses a critical challenge in hydrological science, prompting the exploration of various methodologies. To address this issue, the International Union of Hydrological Sciences (IAHS) launched the “Predictions in Ungauged Basins (PUB)” scientific decade between 2003 and 2012 (Sivapalan et al., 2003). Since then, significant progress has been made in the development of streamflow estimation methods. Singh and Devi (2022) discussed a comprehensive review of the regionalization methods and challenges faced during the pre-PUB (2003–2012) and post-PUB (2013–2021) periods.

Streamflow estimation in ungauged basins is a pivotal endeavor in hydrology, as accurate predictions are crucial for effective water resource management. Conventional approaches play a significant role in addressing this challenge. These methods often involve transferring information from neighboring gauged basins through spatial interpolation or statistical techniques (Ergen & Kentel, 2016; Farmer & Vogel, 2013; Razavi & Coulibaly, 2016; Waseem et al., 2015; Yilmaz & Onoz, 2019, 2020, 2022; Zelelew & Alfredsen, 2014). The drainage area ratio (DAR) method is a widely employed hydrological technique used to estimate streamflow at ungauged sites by extrapolating information from nearby gauged basins (Farmer & Vogel, 2013). Based on the principle that streamflow characteristics are influenced by the size of the contributing drainage area, this method establishes a correlation between the streamflow of a gauged basin

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and its drainage area. Flow duration curves (FDCs) offer insights into temporal flow variability, aiding in predicting streamflow values based on gauged data (Burgan & Aksoy, 2022). Empirical equations establish correlations between streamflow and basin attributes, facilitating straightforward estimations. Physically-based models simulate hydrological processes, integrating factors like precipitation and land properties to project flow behavior. Machine learning algorithms, including artificial neural network (ANN) and random forest, learn from historical observations to predict streamflow dynamics (Vaheddoost et al., 2023a, b). Spatial interpolation methods, such as inverse distance weighting (IDW), extend streamflow information from nearby gauged sites to ungauged areas. Hydrological similarity approaches transfer knowledge from analogous basins to estimate streamflow in similar ungauged locations (Yilmaz & Onoz, 2020, 2022). Recently, conceptual models and machine learning approaches have been proposed as coupled and hybrid models for streamflow estimations (Okkan et al., 2021). These models combine the strengths of both approaches to improve the accuracy and effectiveness of streamflow estimations. By integrating the capabilities of conceptual models in representing physical processes and the flexibility of machine learning algorithms in capturing complex patterns and non-linear relationships, these coupled and hybrid models offer promising solutions for more reliable streamflow estimations.

Streamflow estimation in ungauged basins has experienced a significant advancement with the application of emerging techniques like machine learning, particularly ANN (Besaw et al., 2010; Razavi & Coulibaly, 2013). In the context of ungauged basins, ANN offers a data-driven approach that learns from historical streamflow records in gauged basins. ANN can discern intricate relationships between various hydrological variables, enabling them to learn from neighboring gauged basins and extrapolate predictions for ungauged regions. By analyzing a wide range of influential factors such as precipitation, land use, topography, and soil characteristics, ANN can uncover complex patterns and non-linear interactions that drive streamflow dynamics (Isik et al., 2013; Noori et al., 2014). However, while ANN shows promise, its

successful implementation requires careful consideration of model architecture, input selection, and validation techniques to ensure the reliability and applicability of the estimated streamflow values (Katipoglu et al., 2023; Prasad et al., 2017). When combined with conventional methods such as regionalization or similarity-based approaches, ANN can enrich the predictive capabilities for streamflow estimation (Mohammadi et al., 2021; Noori & Kalin, 2016). By coupling the data-driven strength of ANN with the interpretability and empirical grounding of traditional techniques, hydrologic scientists can develop a more robust and versatile framework.

Razavi and Coulibaly (2016) proposed a multi-modeling approach involving combining neural network and inverse distance methods for improving streamflow estimation in ungauged basins. They used four models, two data-driven and two hydrological but none of the four models alone performed well in all selected basins across Ontario, Canada. In comparison to the individual models, they found that the combined model was more effective and seemed to be more reliable. Vaheddoost et al. (2023a) combined the multivariate adaptive regression spline (MARS) and random forest (RF) methods with the DAR method to increase the reliability and accuracy of streamflow estimates in ungauged sites of the Coruh River Basin. They first estimated daily streamflows in mainstream sites from nearby downstream or upstream stations on the same river by applying the DAR method. They then used the estimated streamflow from the DAR method as input to the MARS and RF models. By combining RF and MARS with DAR, they concluded that the performance of the models improved slightly, while peak values were detected more accurately.

Based on the above brief literature review, this study aims to address the estimation of daily streamflow in basins where streamflow data are poorly gauged or ungauged and therefore challenging to estimate streamflow. As a result, this study suggests a new framework based on a combined modeling approach to reduce the uncertainty in estimating streamflow while the reliability of the method is also tested using a sample basin located in Turkey. The major novelties and contributions of this study are listed as follows:

- (i) To test the performances of two conventional statistical methods individually and their combination for estimating daily streamflow in ungauged basins.
- (ii) To develop an artificial neural network model, based on areal precipitation, streamflow at the selected donor basin, and the estimated streamflow from best-fit statistical methods.
- (iii) To introduce an effective framework that combines statistical methods and artificial neural network to improve the ability to more accurately estimate the daily streamflow.
- (iv) To provide a practical and systematic way of estimating streamflow in ungauged basins.

To achieve this goal, we applied two conventional statistical methods: the drainage area ratio (DAR), and the standardization with mean (SM) which requires regional regression equations for estimating the mean of the annual streamflow as a function of climate and basin characteristics. Then, to improve the estimation performance of the statistical methods, a performance-based ensemble approach combined estimations from the statistical methods into ensemble estimates. The best-fit method was determined among two statistical methods and their combination for each station. Finally, to provide a significant improvement in the final estimation of the ungauged basin, estimated daily streamflow by the best-fit method, daily precipitation for the target station, and daily streamflow of the donor station were used as inputs to ANN for developing the proposed approach.

2. Study Area and Data

This study was carried out in the Coruh River Basin in northeastern Turkey with many major hydroelectric dams under construction and planned, which is an important region in meeting Turkey's energy needs (Fig. 1). The Coruh River which gives its name to the basin originates from the Mescit mountains in Erzurum (above 3000 m above sea level). The bed through which the Coruh River passes is very narrow and deep. The hills and valleys get steeper and the river bed becomes narrow. So, the flow of the Coruh River is very fast. The Coruh River

reaches the highest level of the water regime in spring (April–May), and the lowest water level in winter (December–January). The reasons for the rise in the level in spring are the melting of snow and high seasonal precipitation. The basin is within the borders of three provinces (Artvin, Erzurum, and Bayburt). There is great variability in the climate of the basin. A cold and humid climate is dominant in Artvin while a semi-arid climate is dominant in Erzurum and Bayburt. Temperatures are generally proportional to basin heights. In winter, temperatures are usually around 0 °C or below. On the other hand, temperatures increase in summer and spring. There is typically a decrease in average annual precipitation from the upstream to the downstream of the basin. Natural disasters are one of the most important environmental problems in the study area. Floods are the most common natural disasters in the area and occur twice as often as landslides (GDWM, 2020; Selek & Aksu, 2020).

The selection of streamflow gauging stations and meteorological stations in the basin took into account their availability of sufficient, continuous, and complete data for statistical analysis and modeling. In addition, streamflow gauging stations on the same mainstream, which are not affected by human activities and external factors as much as possible, were preferred. Daily streamflow records (water years 1993–2010) from selected seven streamflow gauging stations on the mainstream of Coruh River were obtained from State Hydraulic Works of Turkey (DSI in Turkish acronym). Streamflow data from the DSI is publicly available and downloadable from <https://www.dsi.gov.tr/Sayfa/Detay/744>, and precipitation data from the Turkish State Meteorological Service (MGM in Turkish acronym) can be purchased at <https://mevbis.mgm.gov.tr/mevbis/ui/index.html>. For the time period covered by the data, water was only utilized for irrigation and public water supplies in the basin. Most dam development projects began after 2010, with the majority of them intended for energy generation. Due to the minimal water usage in the area, naturalization of the streamflow process was not employed in this context. Selected streamflow gauging stations are E23A004, E23A020, E23A038, E23A016, E23A037, E23A005, and E23A035 from upstream to downstream of the Coruh River,

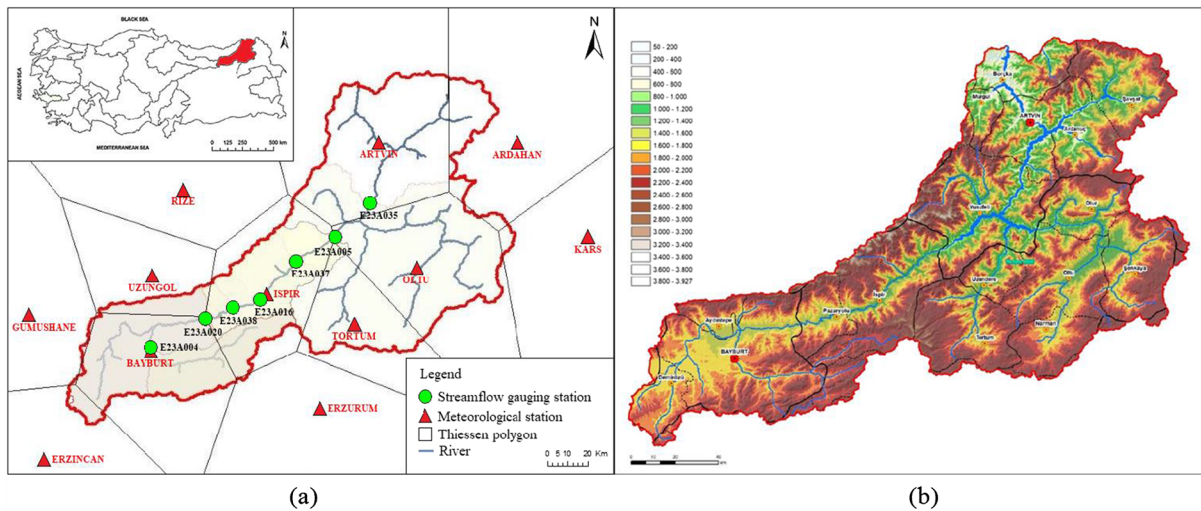


Figure 1

Overview of the study area: **a** location map of selected streamflow gauging and meteorological stations, **b** digital elevation map

Table 1

Some characteristics of selected streamflow gauging stations

Station number	E23A004	E23A005	E23A016	E23A020	E23A035	E23A037	E23A038
River name	Coruh	Coruh	Coruh	Coruh	Coruh	Coruh	Coruh
Station name	Bayburt	Peterek	Ispir	Laleli	Inanli	Camlikaya	Pazaryolu
Lat (°)	40.26	40.74	40.46	40.39	40.89	40.63	40.44
Lon (°)	40.23	41.48	40.96	40.60	41.72	41.21	40.78
E (m)	1545	654	1170	1365	435	892	1265
A (km ²)	1734.0	7272.0	5505.2	4759.2	15,687.1	6634.2	5168.1
Q (m ³ /s)	16.05	71.70	39.53	30.16	122.08	57.57	34.63
P (mm)	480.29	515.67	539.06	538.91	485.88	526.97	538.89
CL (km)	113.32	273.64	234.40	180.83	302.69	243.83	193.35
S (%)	0.0138	0.0088	0.0076	0.0088	0.0082	0.0091	0.0095

A drainage area, *E* elevation, *Q* mean streamflow, *P* total precipitation, *CL* channel length, *S* basin slope, *Lat* latitude, *Lon* longitude

respectively (Fig. 1a). The Thiessen polygon method was used to determine the areal daily precipitation for each station's sub-basin by using meteorological stations in and around the sub-basin. The channel length and basin slope were calculated by using a digital elevation map (Fig. 1b). Several quality control checks were performed on the hydrological and meteorological data.

Information about the selected streamflow gauging stations on the mainstream of the Coruh River is summarized in Table 1. The elevation and drainage areas of the stations range between 435 to 1545 m and between 1734.0 km² to 15,687.2 km²,

respectively. Slopes of the channels are similar for six stations except for one which is the most upstream station (i.e. E23A004). Annual mean total precipitation over the drainage area of the stations varies from 480 to 539 mm. Table 2 presents the statistical characteristics of the observed daily streamflow for the seven streamflow gauging stations. The coefficient of variation for the raw daily streamflow data was between 1.1 and 1.3, which presents less variation of daily streamflow over the basin. The minimum skewness of the coefficient is 2.0, which means highly skewed data with non-normal distribution. The streamflow regime of the basin is similar according to

Table 2

Daily statistical parameters of observed streamflow time series

Station	Obs	Q_{min}	Q_{max}	Q_{mean}	SD	C_s	C_v
E23A004	6574	2.8	232	16.1	19.3	2.9	1.2
E23A005	6574	8.5	646	71.7	81.6	2	1.1
E23A016	6574	3.6	531	39.5	50.3	2.5	1.3
E23A020	6574	1.8	312	30.2	37.9	2.3	1.3
E23A035	6574	14.2	1145	122.1	133.6	2.3	1.1
E23A037	6574	6.4	532	57.6	69.6	2.2	1.2
E23A038	6574	1.6	367	34.6	44.6	2.3	1.3

Obs number of observations, Q_{min} minimum streamflow (m^3/s), Q_{max} maximum streamflow (m^3/s), Q_{mean} mean streamflow (m^3/s), *SD* standard deviation of streamflow (m^3/s), C_s coefficient of skewness, C_v coefficient of variation

the statistical characteristics of the daily streamflow values.

3. Methods

The two statistical methods for daily streamflow estimation in ungauged basins investigated in this study, DAR, SM, and the combination of these statistical methods (i.e. ensemble approach) and the proposed approach described in the following sections were applied to the stations on the mainstream of the Coruh River. The flowchart of the methodology used in the study is illustrated in Fig. 2, where each step is detailed.

3.1. Statistical Methods

3.1.1 Drainage Area Ratio (DAR)

DAR provides the transfer of streamflow data from the nearby gauged basin to the ungauged basin with similar basin characteristics based on the drainage areas of the two stations. Traditionally, the donor station is considered gauged and the target station is ungauged. This method is most suitable for use when the target station is located in the same stream, upstream or downstream of the donor station (Hirsch, 1979; Risley et al., 2008). Its accuracy often depends on the geographical proximity of the two stations. Instead of complex methods that require more input

parameters, it does not require any additional information other than the drainage areas of the donor and the target stations, and the streamflow values of the donor station (Yilmaz & Onoz, 2019). To estimate the streamflow at the target station, multiply the streamflow of the donor station by the ratio of the drainage area of the target station to that of the donor station (Asquith et al., 2006; Emerson et al., 2005). Thus, the estimated streamflow at the target station is given by,

$$Q_{target} = \frac{A_{target}}{A_{donor}} Q_{donor} \quad (1)$$

where Q_{target} is the estimated streamflow at the target station, Q_{donor} is the observed streamflow at the donor station, and A_{target} and A_{donor} are the drainage areas of the target and donor stations, respectively. Equation (1) can be applied to each day to create daily streamflow time series.

3.1.2 Standardization with Mean (SM)

SM is extensively employed in hydrological studies to capture streamflow variability and provide additional hydrologic information not transferred by the ratio of the drainage areas. (Farmer & Vogel, 2013). This method considers the ratio of streamflow to the mean streamflow. If the means of streamflow are known a priori, then the streamflow at the target station can be estimated as,

$$Q_{target} = \frac{\mu_{Q_{target}}}{\mu_{Q_{donor}}} Q_{donor} \quad (2)$$

where $\mu_{Q_{target}}$ and $\mu_{Q_{donor}}$ is the mean of the streamflow at the target and donor stations, respectively. In practice, estimates of the mean streamflow at the target station are required before applying the SM method.

In this study, SM standardizes the daily time series with the mean annual streamflow. The regional regression equations used to determine $\mu_{Q_{target}}$ are based on the climate and basin characteristics. The regional regression equation developed by Farmer and Vogel (2013) for estimating the means of daily streamflow for each target station has the general form of

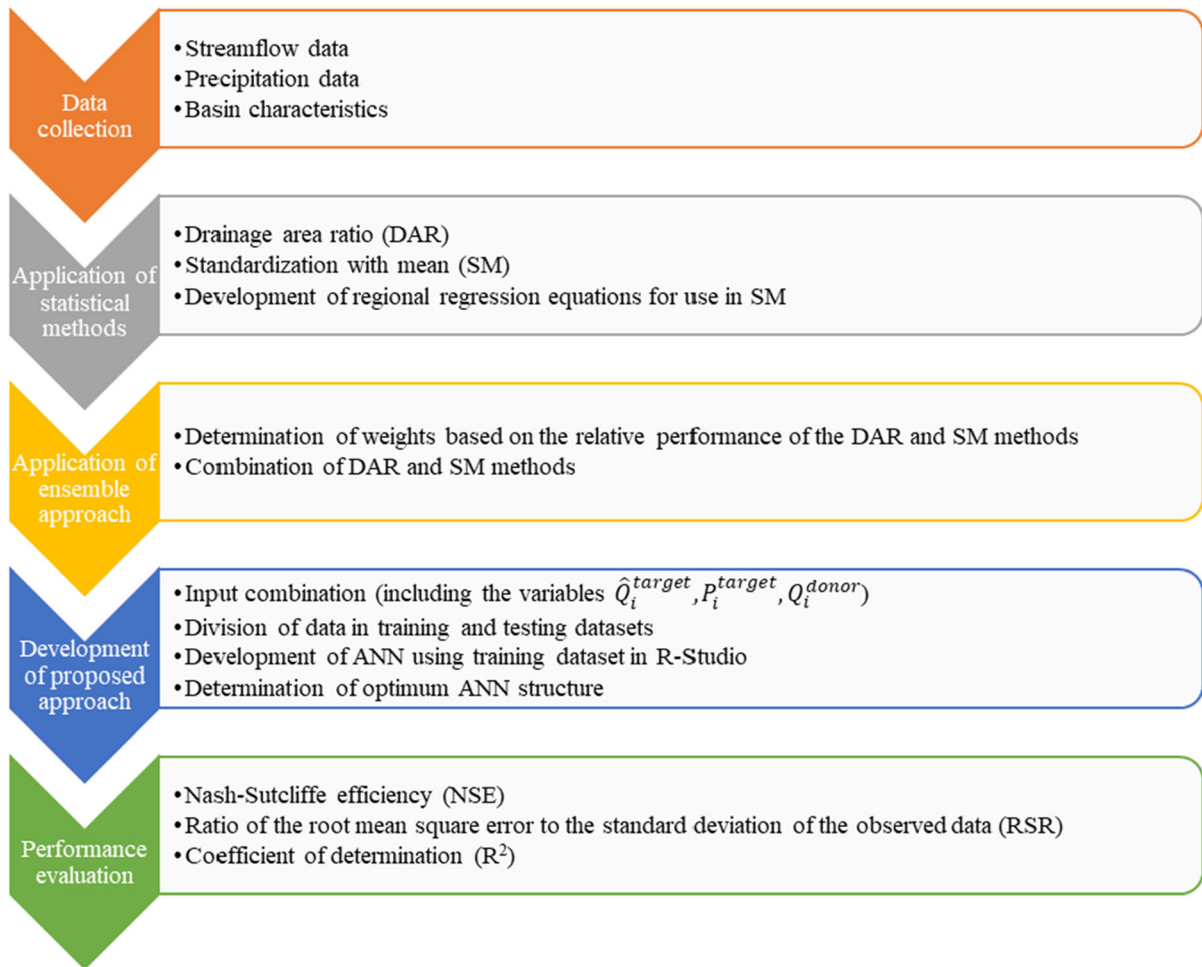


Figure 2
Flowchart of methodology

$$\ln \mu_Q = \beta_0 + \beta_1 \ln X_1 + \beta_2 \ln X_2 + \beta_3 \ln X_3 \dots + \beta_n \ln X_n \quad (3)$$

where μ_Q is the mean of the streamflow; $X_1, X_2, \dots, X_n$ are the climate and basin characteristics; $\beta_0, \beta_1, \beta_2, \dots, \beta_n$ are the regression coefficients.

In this study, the means of streamflow as a function of drainage area, elevation, precipitation, basin slope, and channel length were estimated using regional regressions. A leave-one-out cross-validation procedure was used to test the performance of each regional regression. To develop regional regression equations, data from all stations in the same basin, excluding the target station, was used. Stepwise regression analysis (Agostinelli, 2002) was

applied to obtain the best possible multivariate regression equation for the mean annual streamflow at each target station. This analysis was performed at the 5% significance level and the most significant predictors were determined. Finally, a regression equation was obtained for each target station to estimate the mean annual streamflow.

3.2. Ensemble Approach Based on the Performance of Statistical Methods

The combination of the outputs of the methods in the ensemble approach for an ungauged basin is based on a weighted averaging procedure, which

assigns proper weights proportional to the relative performance of the methods. Such an ensemble approach can provide an appropriate weighting of the two statistical methods that maximize the advantages of each while minimizing their disadvantages. Detailed information on the performance-weighted approach is given by Yilmaz and Onoz (2022). In this study, the ensemble approach proposed by Farmer and Vogel (2013), and Yilmaz and Onoz (2022) which is an NSE-based performance weighted average of DAR and SM methods, was considered. The computation of the ensemble estimate $\hat{Q}_{ens}$ at the ungauged basin from two methods can be obtained using the following equations:

$$\hat{Q}_{ens} = \sum_{m=1}^2 W_m \hat{Q}_m \quad (4)$$

$$W_m = \frac{\frac{1}{(NSE_m - 1)^2}}{\sum_{m=1}^2 \frac{1}{(NSE_m - 1)^2}} \quad \text{and} \quad \sum_{m=1}^2 W_m = 1 \quad (5)$$

where $\hat{Q}_m$ is the daily streamflow estimation from the method m and w_m is the weight assigned to the method m . The values of the weights in Eq. (4) are based on the relative NSE performance of the two methods. The weights range from 0 to 1. The sum of the weights assigned to each method should be equal to 1.

3.3. Proposed Framework Based on Artificial Neural Network and Statistical Methods

This study proposes an effective approach that combines statistical methods and artificial neural network (ANN) to estimate daily streamflow at ungauged basins. ANN is an information processing system that uses interconnected neurons to recreate complex nonlinear input–output relationships (Tongal & Booij, 2018). The daily streamflow of the target station was first estimated with the DAR, SM, and their combination (i.e. ensemble approach), and then the one having the best performance was identified among them, and finally, the estimated streamflow from the best-fit method for each target station was used as inputs to ANN. In addition to the estimated streamflow from the best-fit method, observed streamflow of the donor station and daily

precipitation representing the drainage area of the target station which were considered to have the greatest impact on daily streamflow at the target station were also selected as inputs to ANN. The input structure of the ANN model was found after several experiments and given as follows:

$$Q_i^{target} = f(\hat{Q}_i^{target}, P_i^{target}, Q_i^{donor}) \quad (6)$$

where, $\hat{Q}_i^{target}$ is the estimated daily streamflow at the target station, P_i^{target} is the daily precipitation representing the drainage area of the target station, and Q_i^{donor} is the observed daily streamflow at the donor station.

The entire data set was divided into two subsets. The first about 80% and the last about 20% of data were selected for training and testing of the ANN model, respectively (Table 3). The statistical parameters for each set and the entire data set are presented in Table 4. Training and testing data sets have similar statistical characteristics.

ANN was applied to estimate the daily streamflow for each target station. The optimum number of hidden neurons was determined based on trial and error by varying the number from 1 to 10 for the improvement of the ANN. After trying different numbers of hidden neurons, the optimum number of hidden neurons that gave the best performance on the test data was found for each target station. The network estimation was made using the neuralnet package in R-Studio. Several optimization processes were applied to increase the efficiency of ANN (Tongal & Booij, 2018). The optimized parameters used for ANN are presented in Table 5.

Efficiency in training the ANN can be enhanced through the application of specific preprocessing steps to the network inputs and outputs. Normalization is the most important of these preprocesses. It

Table 3
The training and testing datasets at a daily scale

Period of study	Training period	Testing period
1 Oct 1992 to 30 Sep 2010 (water years 1993 to 2010)	1 Oct 1992 to 30 Sep 2006 78% (5113 days)	1 Oct 2006 to 30 Sep 2010 22% (1461 days)

Table 4

Statistical parameters of training and testing datasets

Station	Period	$Q_{\min}$	$Q_{\max}$	Q_{mean}	SD	C_s	C_v
E23A004	Train	2.8	232	15.8	19.5	2.9	1.2
	Test	3.5	143	16.9	18.7	2.8	1.1
E23A005	Train	8.5	646	69.8	81.5	2	1.2
	Test	14.6	488	78.4	81.9	1.9	1
E23A016	Train	3.6	531	38.4	50.1	2.5	1.3
	Test	7.4	320	43.3	50.9	2.2	1.2
E23A020	Train	1.8	312	29.6	38.3	2.4	1.3
	Test	5.5	249	32	36.3	2.1	1.1
E23A035	Train	14.2	1145	117.8	130.2	2.2	1.1
	Test	33.8	1124	137.2	143.8	2.6	1
E23A037	Train	6.4	532	56.3	69.4	2.2	1.2
	Test	11.7	434	62.1	70	2	1.1
E23A038	Train	1.6	367	33.9	45	2.4	1.3
	Test	6.6	270	37.1	43	2.1	1.2

can be used to scale data over a certain range to minimize bias in neural network. Training a network with the normalized data set can both shorten the training time of the network and increase the efficiency of the network. One of the conventional and most common methods for data normalization available in the literature is minimum–maximum normalization (Malekian & Chitsaz, 2021; Mehr & Safari, 2021). In this study, before the training of the ANN, minimum–maximum normalization was performed and the raw datasets were rescaled to range between 0 and 1.

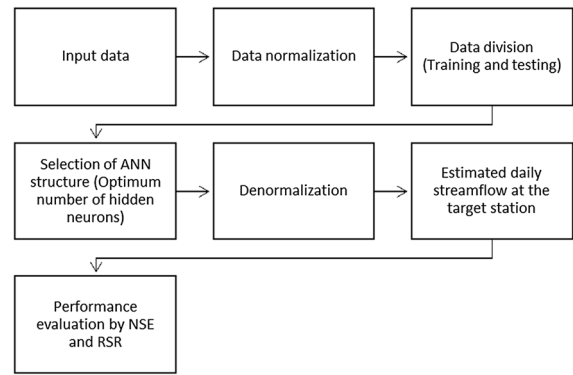


Figure 3
Flowchart of the proposed approach

Before estimating the performance of the approach, all the outputs were denormalized. Various performance measures were used to compare the results estimated by the proposed approach on the training, testing as well as entire data (training and testing). The basic idea behind the concept is to increase the accuracy of the final estimations by using all the data and estimations of the statistical methods as inputs to ANN. The flowchart of the proposed approach is given in Fig. 3.

3.4. Estimation Performance of the Methods

Each target station is in turn considered as being ungauged to estimate streamflow in that basin, the observed streamflow was afterward used to evaluate the performance of the streamflow estimation

Table 5

Training parameters of ANN

Parameters	Optimized values
Algorithm	Resilient backpropagation with weight backtracking (RProp +)
Hidden layer	1
Activation function	Hidden layer Output layer
	Hyperbolic tangent (tanh) Linear
Number of neurons	Input layer Hidden layer Output layer
	3 ^a 1–10 1 ^b
Dataset size	Total Training Testing
	6574 5113 1461

^aEstimated daily streamflow at the target station, daily precipitation representing the drainage area of the target station, and observed daily streamflow at the donor station

^bDaily streamflow at the target station

Table 6

Performance rates of NSE, RSR, and R^2 for streamflow estimations (adapted from Moriasi et al., 2007)

Metrics	Range	Unsatisfactory	Satisfactory	Good	Very good
NSE	$-\infty$ to $+1$	$NSE \leq 0.5$	$0.5 < NSE \leq 0.65$	$0.65 < NSE \leq 0.75$	$0.75 < NSE \leq 1.0$
RSR	0 to $+\infty$	$RSR > 0.7$	$0.6 < RSR \leq 0.7$	$0.5 < RSR \leq 0.6$	$0 \leq RSR \leq 0.5$
R^2	0 to $+1$	$R^2 \leq 0.5$	$0.5 < R^2 \leq 0.65$	$0.65 < R^2 \leq 0.75$	$0.75 < R^2 \leq 1.0$

procedures for the basin. Nash–Sutcliffe efficiency (NSE), the ratio of the root mean square error to the standard deviation of the observed data (RSR), and the coefficient of determination (R^2) were used to check the performance of the methods (Table 6).

NSE, RSR, and R^2 can be calculated using the following equations:

$$NSE = 1 - \frac{\sum_{i=1}^n (Q_i^{obs} - Q_i^{est})^2}{\sum_{i=1}^n (Q_i^{obs} - \bar{Q}^{obs})^2} \quad (7)$$

$$RSR = \frac{\sqrt{\frac{\sum_{i=1}^n (Q_i^{obs} - Q_i^{est})^2}{n}}}{\sqrt{\frac{\sum_{i=1}^n (Q_i^{obs} - \bar{Q}^{obs})^2}{n}}} = \frac{\sqrt{\sum_{i=1}^n (Q_i^{obs} - Q_i^{est})^2}}{\sqrt{\sum_{i=1}^n (Q_i^{obs} - \bar{Q}^{obs})^2}} = \sqrt{1 - NSE} \quad (8)$$

$$R^2 = \left(\frac{\sum_{i=1}^n (Q_i^{obs} - \bar{Q}^{obs})(Q_i^{est} - \bar{Q}^{est})}{\sqrt{\sum_{i=1}^n (Q_i^{obs} - \bar{Q}^{obs})^2} \sqrt{\sum_{i=1}^n (Q_i^{est} - \bar{Q}^{est})^2}} \right)^2 \quad (9)$$

where; Q_i^{obs} is the i th value of the observed daily streamflows, Q_i^{est} is the i th value of the estimated daily streamflows, $\bar{Q}^{obs}$ is the average of all observed daily streamflows, and N is the total number of observations.

4. Results

In this study, statistical methods, their combination, and the proposed framework were employed for seven stations on the mainstream of Coruh River to improve streamflow estimation performance. Three different methods, namely NSE, RSR, and R^2 , were used as performance measures to assess the effectiveness of the applied methods. The results of each modeling approach were given in the following sections.

4.1. Statistical Methods and Ensemble Approach

Firstly, two conventional statistical methods (i.e., DAR and SM) were implemented for daily streamflow estimation of target stations which were

Table 7

Results of stepwise regression analysis used to determine the annual mean streamflow at each target station

Station	Variable(s)	Equation	R^2
E23A004	E	$\ln \mu_Q = 11.941 - 1.174 \ln E$	0.9888
E23A005	A	$\ln \mu_Q = -4.384 + 0.942 \ln A$	0.9564
E23A016	A, P, CL	$\ln \mu_Q = 11.182 + 0.246 \ln A - 2.845 \ln P + 1.549 \ln CL$	0.9996
E23A020	A	$\ln \mu_Q = -4.401 + 0.953 \ln A$	0.9554
E23A035	CL, S	$\ln \mu_Q = -2.879 + 2.260 \ln CL + 1.171 \ln S$	0.9909
E23A037	A, E, CL	$\ln \mu_Q = 3.075 \ln A - 0.682 \ln E + 0.543 \ln CL$	0.9995
E23A038	A	$\ln \mu_Q = -4.504 + 0.964 \ln A$	0.9465

Q annual mean streamflow, A drainage area, E elevation, P precipitation, S basin slope, CL channel length, R^2 coefficient of determination from the regression

Table 8

Performance evaluation of the statistical methods and the ensemble approach at each target station during the water years 1993–2010

Target station	Donor station	NSE			RSR		
		DAR	SM	Ensemble (DAR-SM)	DAR	SM	Ensemble (DAR-SM)
E23A004	E23A020	0.787	– 0.146	0.817	0.461	1.070	0.427
E23A005	E23A035	0.884	0.857	0.874	0.341	0.378	0.355
E23A016	E23A038	0.976	0.910	0.979	0.155	0.301	0.146
E23A020	E23A038	0.982	0.821	0.981	0.135	0.423	0.138
E23A035	E23A005	0.798	0.797	0.968	0.449	0.450	0.179
E23A037	E23A005	0.972	0.966	0.989	0.166	0.185	0.105
E23A038	E23A020	0.984	0.930	0.987	0.125	0.265	0.115

Bold values represent the method with the best performance obtained for each target station. Ensemble approach: the combination of the DAR and SM methods

assumed as ungauged. These methods enabled daily streamflow data to be transferred from a donor station to a target station. For these methods, the donor station is selected as the one geographically closest to the target station, taking into account their hydro morphological similarity. Regional regression was utilized to estimate the mean annual streamflows at each target station for the SM method. In this study, stepwise regressions were used to analyze the independent variables, including drainage area (A), elevation (E), precipitation (P), basin slope (S), and channel length (CL), using data from all stations except for the data of the target station. The regressions were at a 5% confidence level. The results of the regression analysis are summarized in Table 7. The table reports the variable(s), regression equations, and coefficient of determination (R^2). Note that, the regression variables include drainage area for five out of the seven stations. The long-term annual mean of the most upstream station (i.e. E23A004) was estimated by the only elevation while the most downstream one (i.e. E23A035) was estimated by basin slope and channel length. The R^2 values of the equations are significantly high and range between 0.9465 and 0.9995. The incorporation of hydromorphological and climatological variables into regression models improves the model efficiency same as reported in a study conducted for annual streamflow estimation in the USA (Vogel et al., 1999). The results unequivocally illustrate the applicability of utilizing regional regression models with a

stepwise method for estimating long-term annual streamflow within the study area.

DAR and SM methods were combined based on their relative performance to minimize the disadvantages of the methods and exploit their benefits. Both methods perform linear scaling based on the parameter they use (drainage area or average streamflow). DAR and SM methods are combined using an ensemble approach, leading to an enhancement in estimation effectiveness. The weight assigned to each method was determined based on its NSE value.

Table 8 presents the performance evaluation of the statistical methods (DAR and SM) and the ensemble approach for each target station, using NSE and RSR as performance measures. The results demonstrate that the ensemble approach provides a more accurate estimation and a better representation of the hydrological process compared to using the

Table 9

The best result for daily streamflow estimation using the proposed approach

Station	Number of hidden neurons	NSE		RSR		R^2	
		Train	Test	Train	Test	Train	Test
E23A004	1	0.917	0.931	0.289	0.263	0.917	0.932
E23A005	5	0.978	0.975	0.147	0.157	0.978	0.980
E23A016	2	0.983	0.975	0.132	0.159	0.983	0.979
E23A020	2	0.992	0.991	0.091	0.096	0.992	0.991
E23A035	2	0.975	0.956	0.157	0.210	0.975	0.965
E23A037	2	0.992	0.992	0.088	0.089	0.992	0.992
E23A038	3	0.992	0.991	0.090	0.095	0.992	0.991

Table 10

Summary of NSE values for the models at each target station in water year periods 1993 to 2006 and 2007 to 2010

Station	DAR		SM		Ensemble (DAR-SM)		Proposed approach	
	1993–2006	2007–2010	1993–2006	2007–2010	1993–2006	2007–2010	1993–2006	2007–2010
E23A004	0.789	0.778	– 0.135	– 0.187	0.819	0.811	0.917	0.931
E23A005	0.879	0.899	0.852	0.874	0.869	0.890	0.978	0.975
E23A016	0.979	0.966	0.903	0.933	0.981	0.971	0.983	0.975
E23A020	0.983	0.978	0.827	0.796	0.982	0.978	0.992	0.991
E23A035	0.780	0.848	0.816	0.742	0.972	0.955	0.975	0.956
E23A037	0.974	0.967	0.966	0.966	0.990	0.988	0.992	0.992
E23A038	0.985	0.982	0.930	0.929	0.987	0.985	0.992	0.991

Bold values represent the method with the best performance obtained for each target station

statistical methods alone. The ensemble approach significantly improves the streamflow estimation performance for most of the target stations. In contrast, the SM method exhibits the poorest performance for all target stations in the study area. The DAR method outperforms SM for all stations, and it performs best at two out of seven stations among all the methods. This indicates that the DAR method is robust and has good estimation capabilities for the target stations on the same stream.

4.2. Proposed Modeling Framework

In addition to the ensemble approach that combines two conventional methods through linear scaling, an Artificial Neural Network (ANN) was employed to mitigate errors, aligning with the non-linear characteristics of hydrological processes. The proposed modeling framework is a useful approach for estimating daily streamflow in ungauged basins.

This approach was evaluated using various performance metrics, and Table 9 presents all the results for both the training period (1993–2006) and the testing period (2007–2010), along with the determined optimum number of hidden neurons for each target station. The performances during training and testing periods show good agreement with each other. During the training period, NSE values range between 0.917 and 0.992, while during the testing period, NSE values range between 0.931 and 0.991. As shown in Table 9, the proposed approach significantly improves the model's performance across all evaluation measures (NSE, RSR, and R^2) compared to using individual statistical methods. The findings indicate that the coupled framework enhances the accuracy and reliability of streamflow estimation, making it a favorable choice for predicting streamflow at the target stations.

4.3. The Performance Comparison of the Models

NSE values were calculated for both periods (1993–2006) and (2007–2010) to assess the performances of the methods simultaneously. Table 10 presents a summary of the NSE values range for DAR, SM, DAR-SM, and the proposed coupled approach.

Upon comparing the performances of all models for the Coruh River Basin, it is evident that the proposed approach demonstrated the best performance. The NSE values for the proposed approach

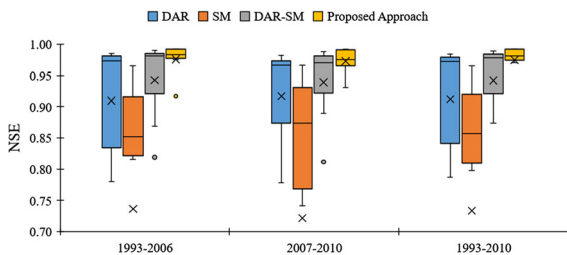


Figure 4

Box plots of NSE values for all target stations using the statistical methods, the ensemble approach, and the proposed approach

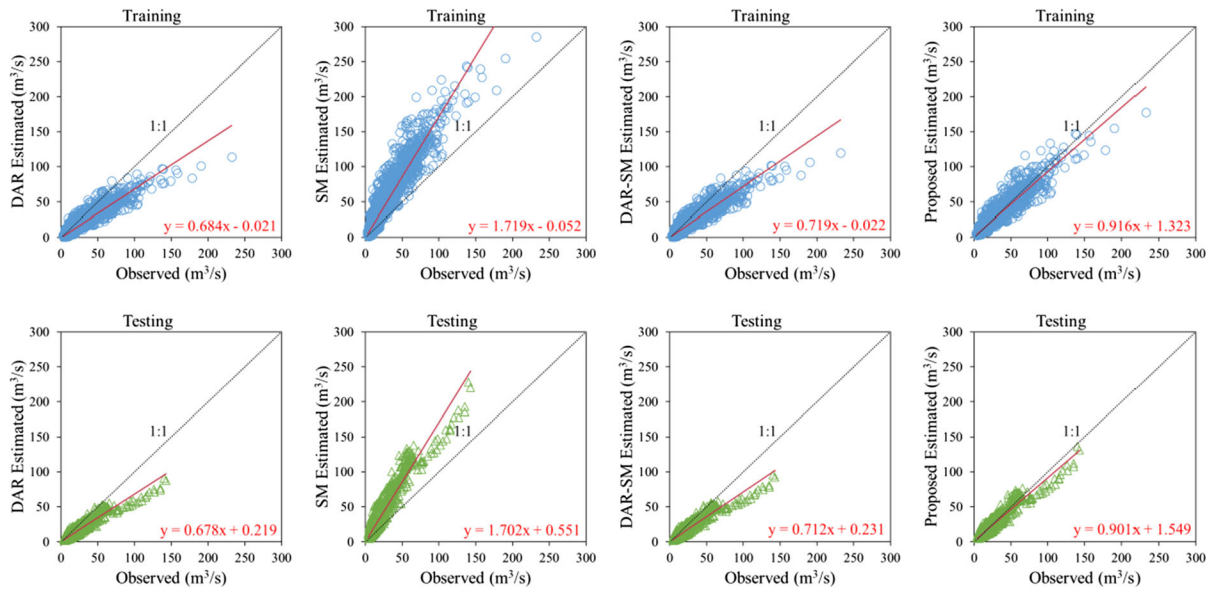


Figure 5

Scatter plots of observed and estimated daily streamflow values of training and testing periods of all methods for station E23A004

are consistently classified as “very good” for all target stations. These results highlight the regional applicability and effectiveness of the proposed approach for streamflow estimation in the study area.

The box plots provide insight into the general behavior of each method across all target stations in the study basin. The accuracy of the methods can be observed from the mean, median, interquartile range (IQR), and minimum and maximum values shown in the box plots. Figure 4 displays the box plots of NSE values for all target stations in the Coruh River Basin using the DAR, SM, ensemble, and coupled approaches for the periods of water years 1993 to 2006, 2007 to 2010, and 1993 to 2010 (entire period). The mean NSE values for all methods were rated as “very good” ($NSE \geq 0.75$) in all periods. However, the variability in the performance of both DAR and SM methods was quite high. The training performance (from 1993 to 2006) showed slightly better results than the testing performance (from 2007 to 2010) and the entire period (from 1993 to 2010).

With the ensemble approach, the variability in performance decreased for all periods. The coupled approach demonstrated the lowest variability, with an IQR of 0.974–0.992 for the entire period. The proposed approach had the highest mean NSE value

(0.975; ranging from 0.920 to 0.992), followed by the ensemble approach (0.942; ranging from 0.817 to 0.989), DAR (0.912; ranging from 0.787 to 0.984), and SM (0.734; ranging from -0.146 to 0.966) for all target stations in the entire period. The application of the coupled approach significantly improved the overall performance of the statistical methods for all target stations.

As an example, Fig. 5 presents scatter plots showing the comparison between observed and estimated daily streamflow values for E23A004 using each method during the training and testing periods. Among the methods, the proposed approach for E23A004 resulted in the closest estimates to the observed daily streamflow values, with NSE values of 0.917 for the training dataset and 0.931 for the testing dataset. The R^2 values were calculated as 0.917 for the training dataset and 0.932 for the testing dataset, indicating a highly linear relationship between the observed and estimated values of daily streamflow. The estimated daily streamflow values showed a strong correlation with the observed values in the proposed approach (i.e., high R^2 values). However, in the other methods, the estimated values deviated significantly from the 1:1 line, resulting in notable overestimation or underestimation. DAR

tended to underestimate the streamflows, while SM tended to overestimate the streamflow at this station. These deviations were particularly noticeable for high streamflow values. The ensemble approach (DAR-SM) showed an improvement in the estimations compared to individual methods, but it still exhibited underestimation for high flows. In contrast, the proposed approach skillfully estimated the streamflow values, making it the proposed approach not only for E23A004 but also for all stations in the study area, considering its high performance.

5. Discussion

Streamflow estimation in ungauged basins stands as a pivotal challenge in hydrology, necessitating innovative approaches at the nexus of data science and hydrological expertise. Conventional methods typically resort to spatial interpolation from neighboring gauged basins, which can fall short of capturing the intricate hydrological processes unique to ungauged areas (Waseem et al., 2015). Conventional methods, while often simple and interpretable, struggle to capture the intricate relationships that define hydrological behavior (Guo et al., 2021). Leveraging artificial neural network, among other techniques, emerges as a promising solution (Hussain et al., 2021). When coupled with conventional methods, ANN can complement the strengths of traditional approaches while mitigating their shortcomings, potentially leading to more accurate and reliable streamflow estimations in ungauged basins (Ersoy et al., 2022; Okkan et al., 2021; Vaheddoost et al., 2023a). The collaboration between these methods introduces the prospect of enhanced hydrological understanding and improved water management strategies for regions where direct observations are lacking.

The current study proposes a new framework coupled with the ANN model to improve the accuracy of flow predictions obtained by traditional statistical methods. This study has yielded successful outcomes in the Çoruh basin and is deemed applicable to other basins with comparable hydrological characteristics. However, there is a decreasing trend in the accuracy of extreme value predictions.

Nonetheless, especially for ungauged basins, there exists potential for improvement by utilizing hybrid and coupled models to enhance streamflow predictions and address data gaps.

Data-driven hydrological studies face two fundamental challenges. First and foremost, data-driven models often exhibit a deficiency in providing adequate physical explanations for hydrological processes (Mohammadi et al., 2022). The second challenge involves the inability of these models to adequately represent factors that influence hydrological processes, such as climate change or urbanization (Okkan & Inan, 2015). Although numerous methods have been proposed in the literature, it is essential to assess various approaches to identify the most suitable and practical one for a specific region (Katipoglu et al., 2023; Yilmaz & Onoz, 2022). Moreover, successful implementation demands a comprehensive understanding of the study area's hydrological characteristics and an effective data collection strategy.

In recent years, several studies have carried out successful applications of both conventional methods and coupled approaches in ungauged basins (Razavi & Coulibaly, 2016; Vaheddoost et al., 2023a, 2023b; Yilmaz & Onoz, 2022). Traditional techniques, which rely on established hydrological equations and empirical relationships, have been effectively utilized to estimate streamflow in areas lacking direct gauge measurements. However, recognizing the limitations of these traditional methods in handling the complexities of ungauged basins, there has been a notable trend toward adopting the coupled method.

In conclusion, the fusion of artificial neural network with hydrological expertise stands to reshape the landscape of streamflow estimation in ungauged basins, fostering a more comprehensive understanding of water availability and enhancing sustainable water resource management.

6. Conclusions

In this study, to estimate more accurately daily streamflow in ungauged basins, we introduced an effective framework, based on the statistical methods and ANN, using the Coruh River Basin as a case study. The proposed approach showed the best

performance among the DAR, SM, and combination of DAR and SM methods for all the stations. The second best method was the ensemble approach and the third is the DAR method. SM method alone did not present the best estimations for any of the models in the study area. All the models tend to perform poorest during the high flows, however, the best performance still belongs to the proposed approach for daily high streamflow estimations. The proposed approach applies the linear and nonlinear approaches together to reflect the complex interactions between hydrological processes. In addition, the regional performance of the proposed approach is significantly higher than the DAR, SM, and their ensemble approach. Meanwhile, using precipitation and estimated streamflow of the target basin, observed streamflow of donor basin as inputs to the ANN, and the proposed approach has a significant effect in estimating daily streamflow at the target basin. The results have shown that the proposed approach is a better effective tool for daily streamflow estimation in ungauged basins. However, the most used regionalization methods are DAR and SM methods respectively, newly introduced ensemble and the proposed approaches have great potential to improve the streamflow estimations. It is important to evaluate the performances of streamflow estimations methods to determine the best methodology considering the diverse climatological and hydro morphological characteristics of the basins.

The proposed approach for daily streamflow estimation can be applied to any other region. For future studies, long-term data records and an intense gauging network may improve the developed methodology. While streamflow estimation in ungauged basins remains still a difficult task over the last two decades, more scientific interest is needed to estimate streamflows in ungauged basins.

In brief, the integration of ANN offers a promising approach for streamflow estimation in ungauged basins. Optimizing the coupling, enhancing the interaction between ANN and traditional methods, and exploring additional hydrological data integration is crucial. Sensitivity analysis and validation techniques can enhance reliability amidst uncertainties. Embracing these suggestions leads to accurate

streamflow predictions, ultimately assisting water resource management in regions with limited data.

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Data Availability

The data that support the findings of this study are available from the General Directorate of State Hydraulic Works of Turkey and the Turkish State Meteorological Service upon official request.

Code Availability

Not applicable for this study.

Declarations

Conflict of Interest The authors declare no conflicts of interest.

Ethical Approval Was not required for this study.

Consent to Participate Was not required for this study.

Consent to Publish Was not required for this study.

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